

Deepak Mallampati

APPLIED AI / AGENTIC SYSTEMS ENGINEER

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US-based, F-1 OPT (open to sponsorship) - Remote

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years designing autonomous agentic systems and production LLM applications. Build agent-orchestration frameworks, RAG pipelines, and evaluation harnesses end-to-end - from design through cloud-native deployment with CI/CD. Shipped a conductor-subagent agentic architecture that cut token consumption 42% while holding accuracy across multi-step workflows, and high-precision RAG over large enterprise corpora with measurable hallucination reduction. Strong on Responsible AI and compliance (HIPAA, SOC 2, content safety), with proven collaboration across product, data science, and compliance teams.

CORE COMPETENCIES

Agentic Systems & Multi-Agent Orchestration • LangChain / LangGraph • RAG Pipelines & Vector Stores • LLM Evaluation (RAGAS, BERTScore) • Fine-tuning (LoRA / QLoRA) • Cloud-Native Deployment & CI/CD • Responsible AI & Compliance

Generative AI & Agents: GPT-4o/GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | Multi-Agent & Autonomous Systems | RAG Architecture | Prompt Engineering | Fine-tuning (LoRA, QLoRA, PEFT) | Hugging Face Transformers

Retrieval & Evaluation: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Document Chunking | RAGAS, BERTScore | Drift Detection | A/B Testing

MLOps & Cloud: AWS (Bedrock, SageMaker, Lambda, S3), Azure (OpenAI, AI Services, Content Safety), GCP Vertex AI | Docker, Kubernetes | Terraform | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | Model Monitoring

Responsible AI & Compliance: HIPAA, SOC 2, PCI DSS | PII Masking/Redaction | Prompt Injection Mitigation | Audit Logging | Content Safety

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI, React | PostgreSQL, MongoDB, Redis | Git, Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present | USA

AI Engineer

- Architected production-grade **autonomous agentic systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents that execute autonomously - reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large-scale document corpora, combining dense vector retrieval with cross-encoder re-ranking and document chunking to lift answer relevance and materially reduce hallucination on domain-specific queries.
- Established a structured **LLM evaluation framework** (RAGAS, BERTScore, custom metrics) with latency and accuracy dashboards to benchmark iterations pre-deployment and surface regressions before release - monitoring accuracy, latency, and cost.
- Optimized inference cost and latency via prompt compression, semantic caching, and model routing, sustaining SLA targets while reducing per-query overhead.
- Built fault-tolerant automation infrastructure with scheduled orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**; deployed cloud-native with CI/CD.

Vanguard

Jan 2024 - Jan 2025 | USA

AI Engineer Intern

- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines, databases, and observability tooling across **25+ AI agents** and workflows.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** and informing platform model-selection decisions.
- Engineered **Responsible-AI safeguards** and content controls for secure, policy-compliant responses, reducing unsafe outputs by **42%** while maintaining quality.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms**; partnered with product, data science, and compliance teams to serve **100,000+ requests daily** across 20+ teams.

Kent State University

Oct 2023 - Jan 2024 | USA

ML & Gen AI Research Assistant

- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 with a curated instruction dataset, conditioning the model on a structured narrative schema for controllable long-form generation.
- Built a **privacy-preserving ML pipeline** (k-anonymity, l-diversity, differential privacy), reducing re-identification risk from **3.38% to 0.32%** while retaining **99%** of baseline predictive performance.

Emerson

Jun 2022 - Apr 2023 | India

ML Trainee

- Trained supervised **regression and classification models** on operational data for equipment-failure prediction and anomaly detection, building end-to-end model-ready data pipelines.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1** on custom NER tasks.

PROJECTS

career-ops

An autonomous, AI-driven pipeline of agents that scans, scores, and generates tailored outputs overnight without supervision - agent orchestration, evaluation, and automated rendering end-to-end.

Story Director LLM

A fine-tuned (QLoRA) language model that generates structured narratives from a single prompt, paired with an automated rendering pipeline - domain-specific fine-tuning applied to a real generation task.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - multimodal LLM integration in a production consumer product.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0